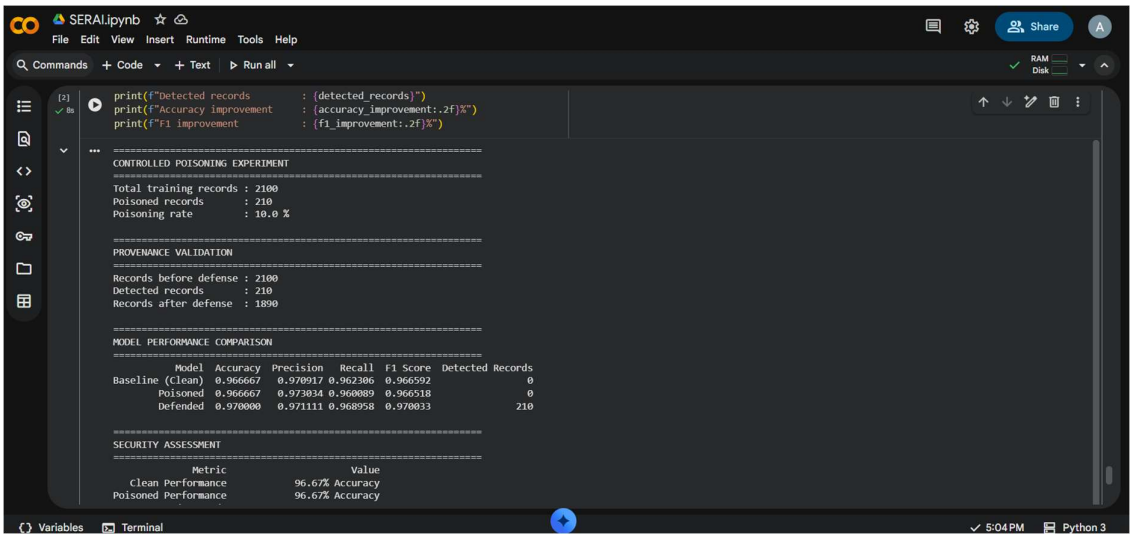
Assignment 4

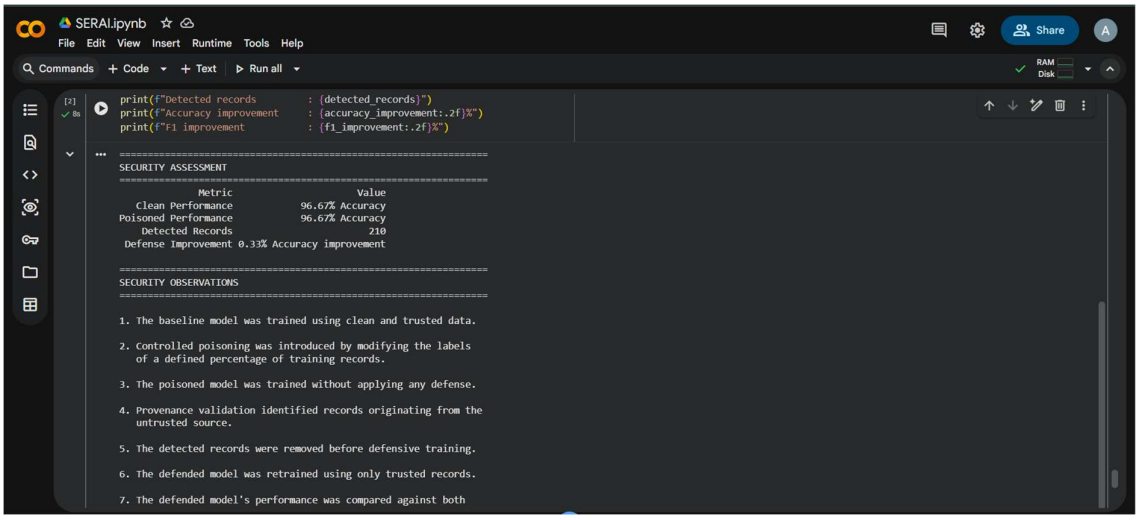
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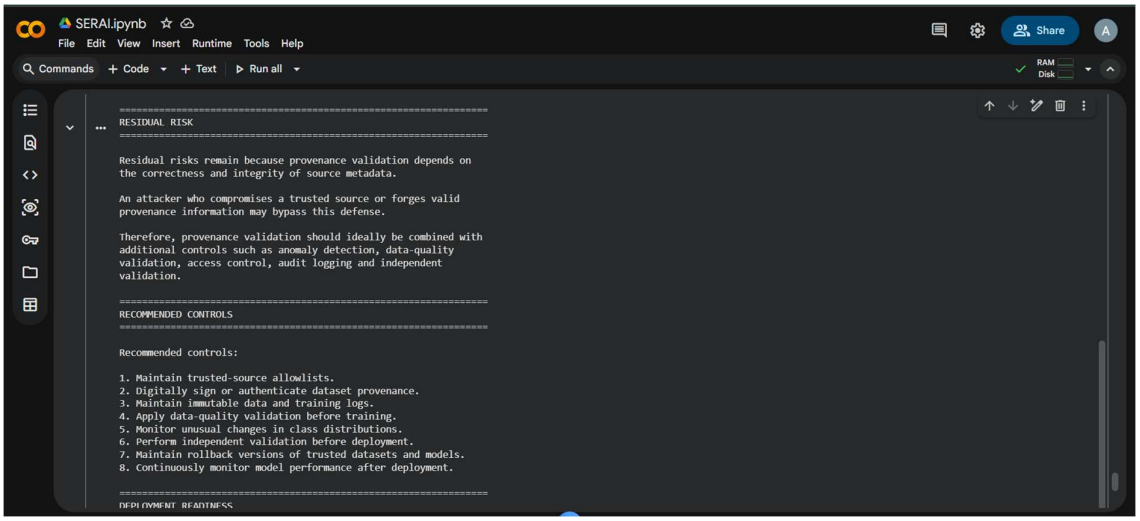
Batch: 23

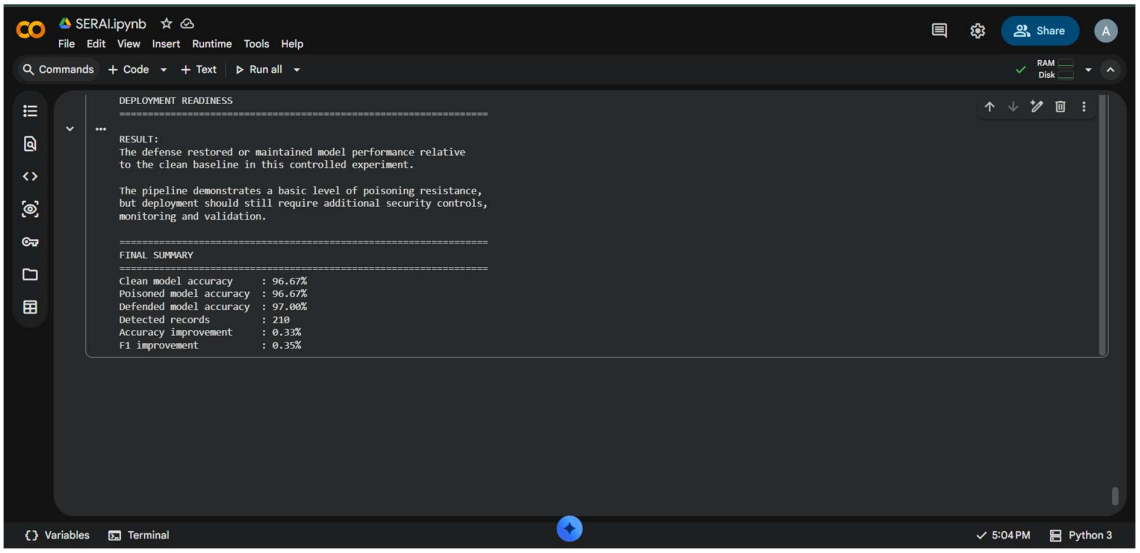
Poisoning Defense and Integrated Security Assessment

Output:






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==
RESIDUAL RISK
==
Residual risks remain because provenance validation depends on
the correctness and integrity of source metadata.
An attacker who compromises a trusted source or forges valid
provenance information may bypass this defense.
Therefore, provenance validation should ideally be combined with
additional controls such as anomaly detection, data-quality
validation, access control, audit logging and independent
validation.
==
RECOMMENDED CONTROLS
==
Recommended controls:
1. Maintain trusted-source allowlists.
2. Digitally sign or authenticate dataset provenance.
3. Maintain immutable data and training logs.
4. Apply data-quality validation before training.
5. Monitor unusual changes in class distributions.
6. Perform independent validation before deployment.
7. Maintain rollback versions of trusted datasets and models.
8. Continuously monitor model performance after deployment.
==
DEPLOYMENT READINESS


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DEPLOYMENT READINESS
==
RESULT:
The defense restored or maintained model performance relative
to the clean baseline in this controlled experiment.
The pipeline demonstrates a basic level of poisoning resistance,
but deployment should still require additional security controls,
monitoring and validation.
==
FINAL SUMMARY
==
Clean model accuracy : 96.67%
Poisoned model accuracy : 96.67%
Defended model accuracy : 97.00%
Detected records : 210
Accuracy improvement : 0.33%
F1 improvement : 0.35%
Variables
Terminal
5:04 PM
Python 3